Tiny Think

Reasoning-First Post-Training for Tiny Math and STEM Language Models

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Why Tiny Think?

- The motivating question was straightforward:
  - can we build **specialized tiny reasoning models** on hardware that normal people can run locally?
- The setup is intentionally constrained:
  - **no cluster**
  - **no multi-GPU training**
  - **no expensive infrastructure**
- Our goal is **not** to claim that 140M models have solved reasoning.
- Our goal is to understand which post-training recipe works best for **math and STEM specialization under real constraints**.

140M
base model parameters

RTX 5060 Ti
single 16 GB consumer GPU

~60M
SFT tokens with think traces

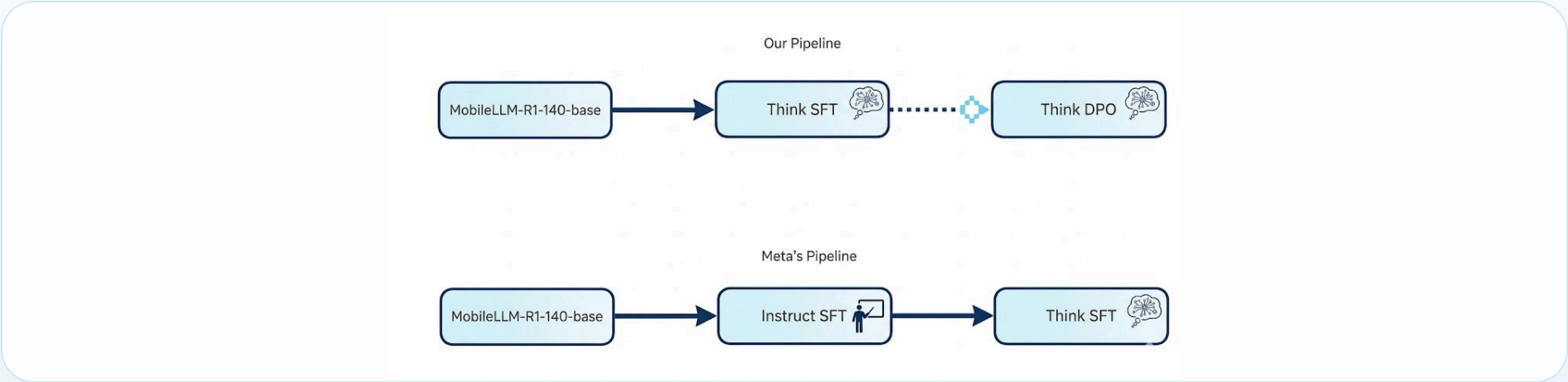
~10M
preference tokens for Stage B

Full FT
no LoRA or distributed setup

Open
code, configs, checkpoints released

Lean Pipeline, Real Constraints

- Start from `MobileLLM-R1-140M-base`
- Run **one reasoning-focused SFT stage** on math + STEM data with explicit think traces
- Then apply **preference optimization** on top
- Inspired by the OLMo 3 style of reasoning-first post-training
- Compared with Meta’s instruct-SFT then think-SFT path, this recipe is leaner and easier to reproduce



What The Training Data Looks Like

SFT — SUPERVISED REASONING TRACE

Prompt: Joel collected a bin of old toys to donate. By asking
his friends he collected 18 stuffed animals, 42 action figures,
2 board games, and 13 puzzles. His sister gave him some toys,
and Joel added twice as many from his own closet. In all, 108
toys. How many of the donated toys were Joel’s?
Response:
<think>Let T be the number of toys Joel’s sister donated.
Joel donated twice as many, so 2T toys.
Friends donated 18+42+2+13 = 75 toys.
T + 2T + 75 = 108 → 3T = 33 → T = 11.
Thus 2T = 22 toys were Joel’s.</think>
Answer is 22

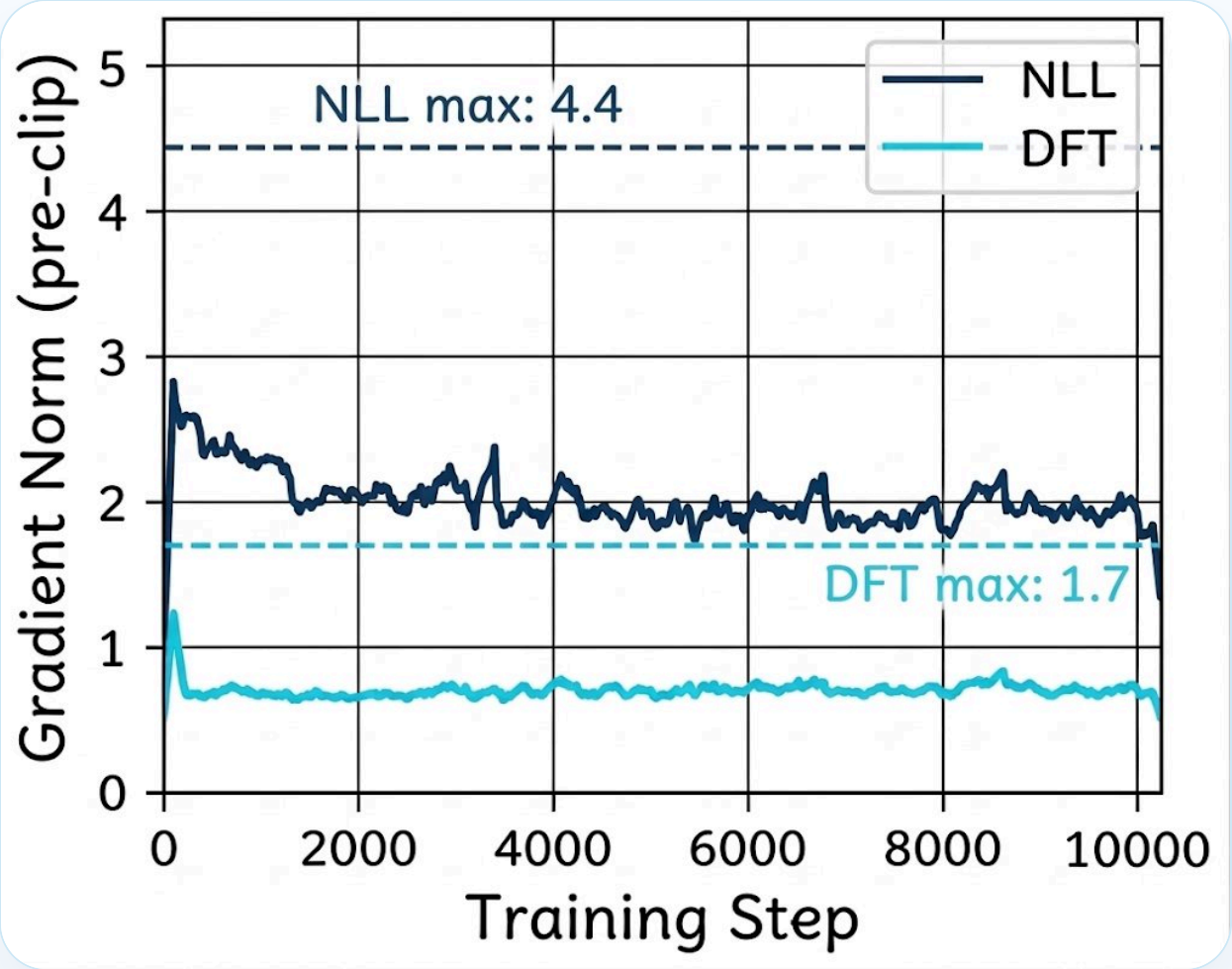
DPO — CHOSEN VS. REJECTED PAIR

Prompt: (same Joel toys problem)
Chosen:
<think>Let T be the number of toys Joel’s sister donated.
Joel donated twice as many, so 2T toys.
Friends donated 18+42+2+13 = 75 toys.
T + 2T + 75 = 108 → 3T = 33 → T = 11.
Thus 2T = 22 toys were Joel’s.</think>
Answer is 22

Rejected:
Joel’s friends donated about 75 toys. Joel and his sister
added the rest. Since Joel gave twice as much, he donated
around half of the remaining 33, so roughly 16 or 17 toys.
Answer is 17

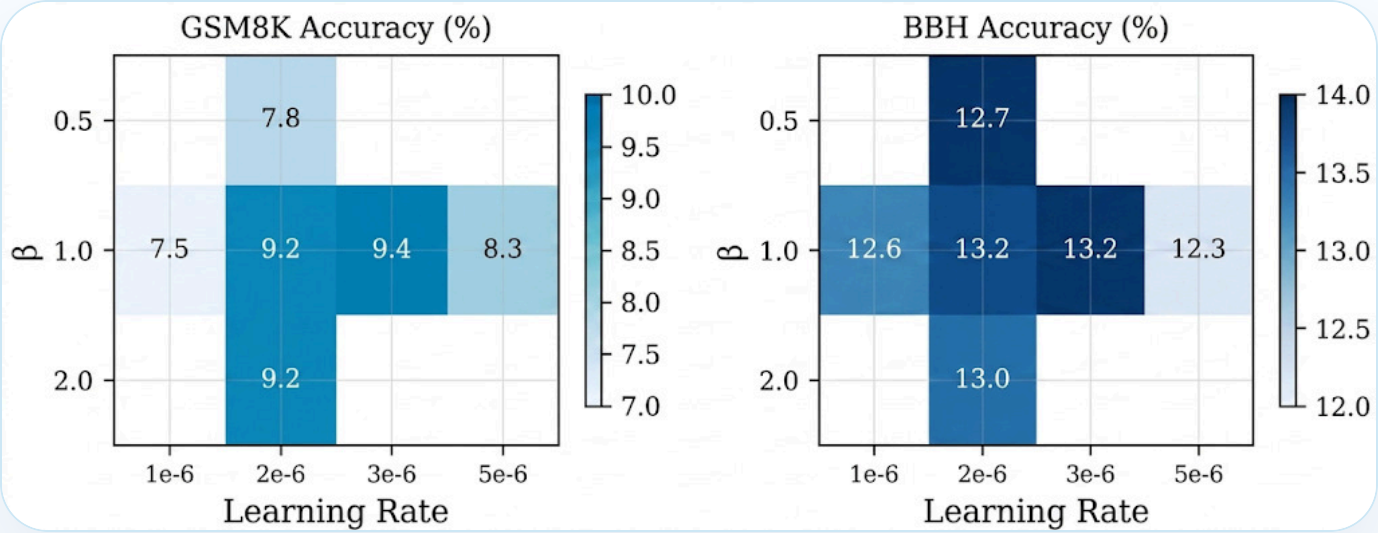
Stage A: DFT Makes SFT Much More Stable

- In SFT, **DFT is much more stable** than standard NLL.
- Peak gradient norm drops from about **4.4 with NLL loss**, down to about **1.7 with DFT**.
- Token accuracy remains in the same broad range.
- But stability is **not the whole story**:
  - **NLL** gives the best math SFT checkpoint
  - **DFT** gives a stronger balanced reasoning/STEM profile



Stage B: Preference Tuning Is Brittle

- Higher learning rates create **larger reward margins**, but those diagnostics do **not** map cleanly to better downstream quality.
- The best GSM8K checkpoint appears at **DPO with beta = 1 and LR = 3e-6**.
- That same hyperparameter region still has **substantially weaker BBH** than the strongest SFT checkpoints.
- So at this scale, preference tuning is **sensitive, high-variance, and easy to over-interpret**.



Main Result: Better Math, Worse General Reasoning

Checkpoint	GSM8K	MATH500	BBH	IFEval
SFT (NLL)	8.04	**4.00**	**23.84**	**21.63**
DPO	**9.40**	1.80	13.18	16.45
APO	8.26	3.80	12.01	16.08

If we evaluate only GSM8K, DPO looks like the winner. If we evaluate the full suite, it is a tradeoff rather than a global improvement.

Interpretation

- **DPO is best on GSM8K only.**
- Relative to the strongest SFT checkpoint, it gives up roughly:
  - **10.66 pp** on BBH
  - **5.18 pp** on IFEval
  - **2.20 pp** on MATH500
- Math-only evaluation would have hidden it.

How Does 9.40% Compare?

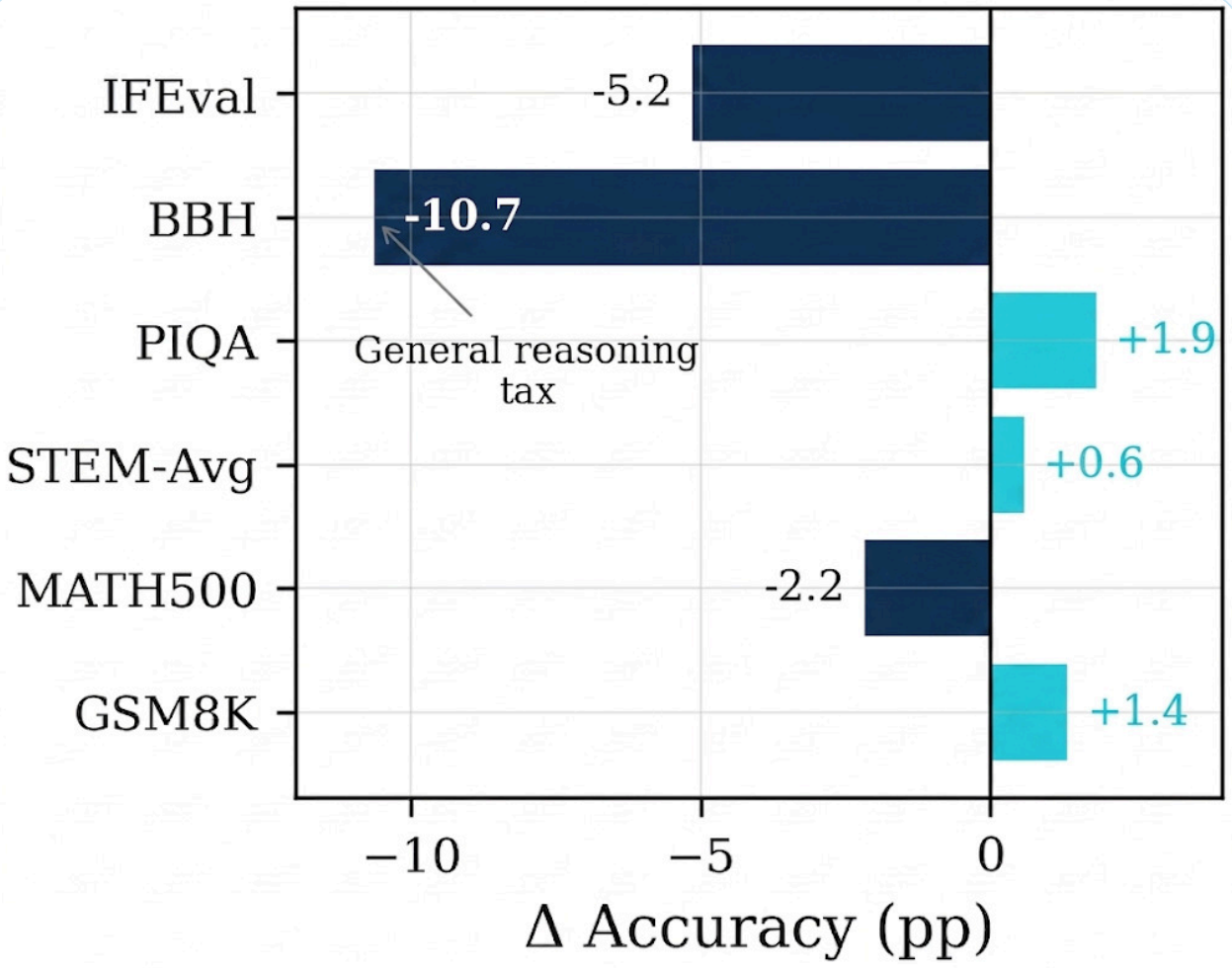
Model	Size	GSM8K
SmolLM2-135M-Instruct	135M	2.40
MobileLLM-R1-140M	140M	4.10
Gemma-3-270m-it	270M	8.40
Tiny Think DPO	**140M**	**9.40**

A 140M model with this recipe outperforms Gemma-3 at 2× the parameters — on a single consumer GPU with no LoRA.

Important caveats

- The contribution here is the **methodology and analysis**, not a leaderboard claim
- This result shows the recipe is **competitive at tiny scale** under strict resource constraints

The General Reasoning Tax



Incremental effect of preference tuning. DPO adds 1.36 points on GSM8K but sharply hurts BBH and IFEval.

Why does this happen?

- **Format specialization** — DPO sharpens toward verbose step-by-step derivations that don't transfer to other task structures
- **Capacity constraints** — at 140M params, optimizing hard toward one domain crowds out others
- **Domain narrowing** — math+STEM-heavy preference data overemphasizes structured reasoning at the cost of general flexibility

At 140M scale, preference optimization is better understood as capability steering or calibration than capability addition.

Takeaways and Next Steps

Core takeaways

1. **Tiny-model reasoning is tractable.** A 140M model can gain meaningful math behavior with a lightweight two-stage recipe.
2. **DFT improves optimization stability.** That matters when compute budgets are tight.
3. **Preference tuning is not a free lunch.** The gains on GSM8K come with measurable regressions elsewhere.
4. **Evaluation discipline is essential.** Multi-objective evaluation is the only reason the general reasoning tax is visible.

What comes next

- Distillation is a natural next step for pushing tiny reasoning further.
- Tiny Think is a good testbed because the recipe is already lean and reproducible.
- The project is also being released openly to keep tiny-model development accessible.

Limitations

- Scope limited to math + STEM; coding, multilingual, safety untested

Closing Thought

tiny-think

Thank you. Code and artifacts:

GitHub

Hugging Face